

We are going to use basic Web technologies:

- HTML for the entry point of our extension
- JavaScript for the logic

Setting up the development environment

Let us set up a file structure for our extension, and create a new folder on the computer. In our case, `C:\wikisearch` is the extension's folder. Now, inside that new folder, create a folder called *images*.

In the root directory of our extension folder, create an empty text file, called *manifest.json* and create an empty HTML file called *background.html*. Also create an empty JavaScript file that you can call *logic.js*.

You should end up with the following directory structure:

- *images*
- *background.html*
- *manifest.json*
- *logic.js*

Description

images is the directory containing icons or images which we will be using in our extension.

background.html is the entry point of our extension. The logic for the extension will be loaded from here.

manifest.json is a metadata file in the JSON format that contains basic properties like the extension's name, description, version number, etc.

logic.js is a JavaScript file containing the logic for the extension.

Adding icons or images for our extension

Here, we are going to add an icon just to make our option in the context menu more appealing. Go to <http://www.flaticon.com/search/magnifier> and download any one icon, preferably in the .png format of size 16x16, and copy it to the images folder located in our extension's folder. Also rename that image to 'search.png'.

Creating the manifest file

Open the *manifest.json* file and copy the following JSON code:

```
{
  "manifest_version": 2,
  "name": "Wiki Search",
  "description": "Searches for the selected text on
Wikipedia.org",
  "version": "1.0",
  "minimum_chrome_version": "20",
  "permissions": [
    "contextMenus",
    "tabs",
    "icons": {
      "16": "images/search.png"
    }
  ]
}
```

```
},
"background": { "page": "background.html" }
}
```

This code has been written in JSON format. It consists of general information about the extension such as its name, description, version, minimum version of the Chrome browser it supports, icons/images used in the extension and so on.

Parameter description

- 'manifest_version' specifies the version of the manifest; the value must be 2.
- 'name' specifies the name of the extension.
- 'description' specifies a short description of the extension.
- 'version' specifies the version of the extension.
- 'minimum_chrome_version' specifies the minimum version of the Chrome browser required to run the extension.
- 'permission' specifies which Chrome APIs are to be used. We are using contextMenus and tabs.
- 'icons' specifies the location of the icons/images we are going to use in our extension.
- 'background' specifies that the page 'background.html' should load as soon as the user starts the Chrome browser. Basically, this will be the entry point for our extension.

Adding JavaScript

Open the *logic.js* file located in our extension directory, and add the following JavaScript code:

```
searchWikipedia = function(word){
  var query = word.selectionText;
  chrome.tabs.create({url: "https://en.wikipedia.org/wiki/" +
query});
};

chrome.contextMenus.create({
  title: "Wiki Search",
  contexts:["selection"],
  onclick: searchWikipedia
});
```

The 'searchWikipedia' function gets the value of the element 'word.selectionText', and stores it in the variable named *query*. Now, using DOM, we just append the extracted value with the URL of the Wikipedia search engine to load the page with query search results.

The next task is to create the UI for our extension, since we are providing the option in the context menu. We will be using the *contextMenus.create()* function to create customised context menu. This function has some key:value parameters such as the title, contexts, and the single most important event – which specifies what must be done when certain browser based events occur (i.e., in our case, the *searchWikipedia()* function will be called).